جامعة عفت

EFFAT UNIVERSITY

CS 4084 - IoT and and Smart Cities

Project

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1 Project Idea

Project Title:

Smart Garbage Bin Monitoring System Using IoT in Saudi Arabia

Project Description

Idea Overview

This project involves creating a smart garbage bin using Arduino, servo motor, and sensors. It opens automatically using a servo motor when someone approaches (detected by an ultrasonic sensor), .The goal is to increase hygiene and optimize waste management without requiring physical contact.

Background and Problem Statement

In public areas of Saudi Arabia, hygiene is essential. Traditional bins require contact to open and do not notify when they are full. This project offers a hands-free, smart alternative that improves cleanliness and efficiency.

Project Significance

The Smart Garbage Bin Monitoring System plays a vital role in supporting clean, sustainable, and smart urban environments. It aligns with Saudi Arabia's Vision 2030 by promoting public hygiene, reducing waste overflow, and enabling smart infrastructure. This automation reduces human interaction with waste and can lead to smarter collection scheduling in public areas.

Why This Project?

This project addresses the real-world need for more hygienic and efficient waste disposal methods. By automating lid opening and monitoring fill level, the project enhances public health and supports smart city services.

How the Project Relates to IoT

The project exemplifies a typical IoT system:

- **Sensing:** It uses ultrasonic sensors to measure distance.
- **Actuation:** It uses a servo motor to physically open or close the bin.
- **Data Collection:** It gathers real-time environmental data (e.g., distance).
- **Future Extension:** This system can be extended to send data to a cloud server for centralized waste management.

Objectives

- Develop an Arduino-powered smart bin.
- Detect presence and open lid automatically.
- Display data and prepare for possible future upgrades like cloud integration.

Components Used

Name	Quantity	Component
Arduino UNO	1	Microcontroller Board
Ultrasonic Sensor (HC-SR04)	1	Distance Measurement
Servo Motor (SG360)	1	Lid Control

Table 1: List of Components Used in the Project

Software and Tools

- **Arduino IDE** – Code development.
- **Tinkercad** – Circuit simulation.

Simulation Details

The circuit was built and tested using Tinkercad to simulate real-time interactions and validate functionality:

- Arduino Uno R3 manages sensor readings and output actions.
- Servo Motor opens/closes the lid.
- Ultrasonic Sensor measures distance.

Simulation Environment (Tinkercad): Used to virtually test the system logic, circuit behavior, and component interaction before building the physical prototype.

Data Collection and Expected Output

- **Distance (cm)** – From the ultrasonic sensor.
- **User Detection** – Determines if lid should open.

2 Project Simulation:

Required Tools and Technologies

1. Hardware Requirements

- Arduino UNO
- Ultrasonic Sensor (HC-SR04)
- Servo Motor

2. Software and Development Tools:

- Arduino IDE – Used to program and upload code to the Arduino.
- Tinkercad Circuits – Online platform to simulate the Arduino circuit and test behavior.

Sample Code (Arduino)

```

#include <Servo.h>

Servo myServo;
int servoPin = 9;

void setup() {
  myServo.attach(servoPin);
}

void loop() {
  myServo.write(0);
  delay(1000);
  myServo.write(90);
  delay(1000);
  myServo.write(180);
  delay(1000);
}
}
```

Listing 1: Arduino Code for Smart Dustbin Simulation

Sample Code (Arduino)

```
const int trigPin = 10;
const int echoPin = 11;

void setup() {
  Serial.begin(9600);
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
}

void loop() {
  long duration;
  float distance;

  // Clear the trigPin
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);

  // Send a 10us HIGH pulse to trigger the sensor
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);

  // Read the echoPin, returns the sound wave travel time in
   microseconds
  duration = pulseIn(echoPin, HIGH);

  // Calculate the distance in cm
  distance = duration * 0.034 / 2;

  // Print the distance to Serial Monitor
  Serial.print("Distance: ");
  Serial.print(distance);
  Serial.println(" cm");

  delay(1000); // wait 1 second before next reading
}
```

Listing 2: Arduino Code for Smart Dustbin Simulation

Data Used / Collected in Project Simulation

During the simulation and testing of the smart garbage bin system, the following data is collected in real time:

- **Distance (cm)** – Captured using the HC-SR04 ultrasonic sensor to measure the proximity of a user or object to the bin.
- **Lid Status** – Logged based on whether the bin lid is open or closed, determined by distance readings.

This data can be used for decision-making (e.g., opening/closing the lid) and can be extended in future versions to support IoT-based cloud monitoring or smart city dashboards.

Simulation Results / Screenshots

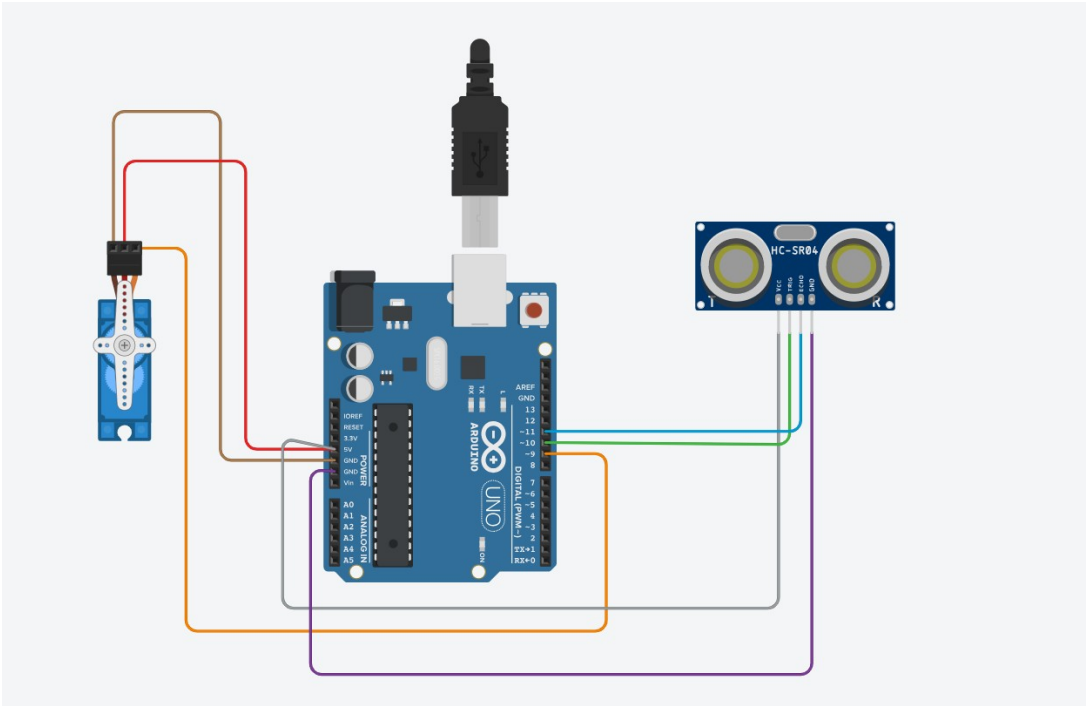

Figure 1: Circuit Diagram -1

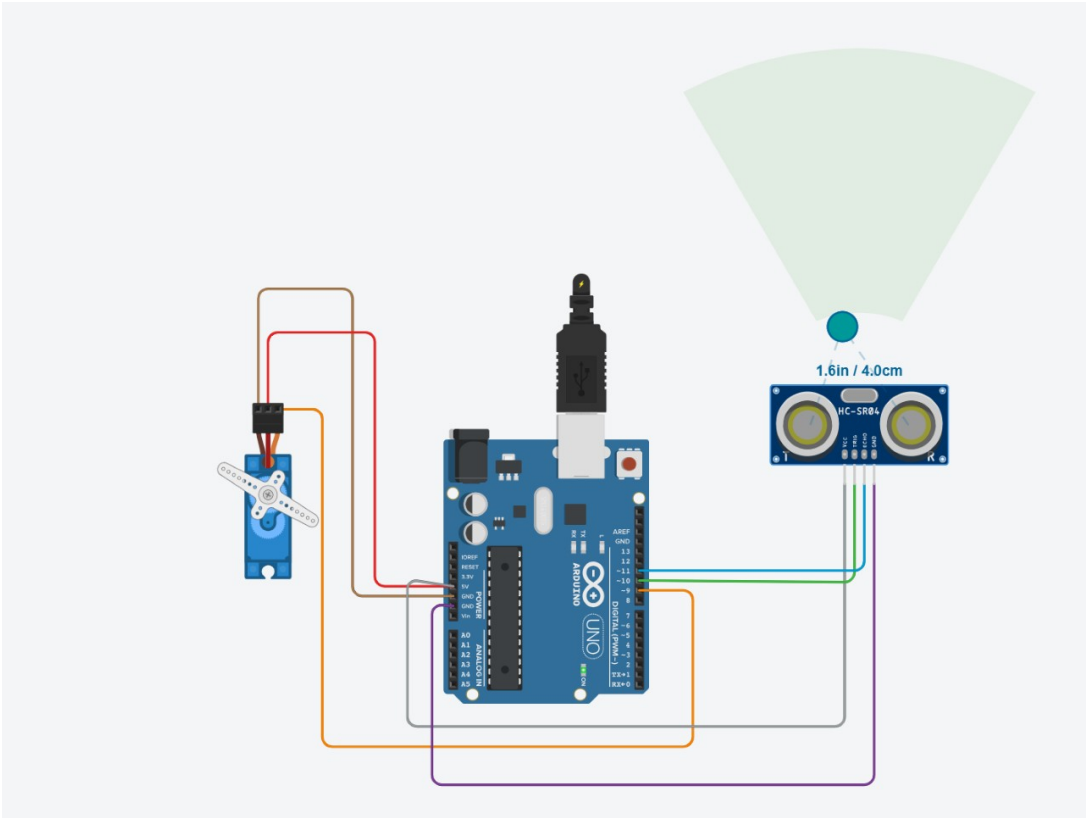

Figure 2: Circuit Diagram -2

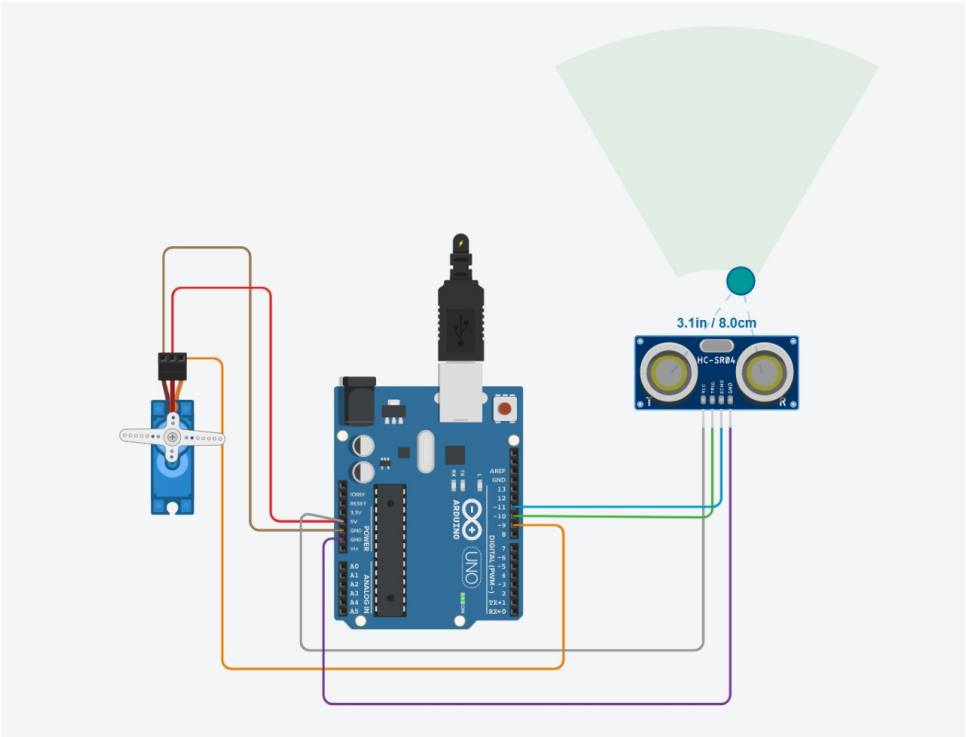

Figure 3: Circuit Diagram -3

Figure 4: Prototype1

Figure 5: Prototype2

Table 2: Pin Connections for Smart Dustbin Project

Component	Pin	Arduino Connection
Ultrasonic Sensor (HC-SR04)	VCC	3.3V
	GND	GND
	TRIG	D10
	ECHO	D11
Servo Motor (SG90)	VCC	5V
	GND	GND
	Signal	D9

Table 2: Pin Mapping for Sensor and Actuator Components

3 Arduino Code

```
#include <Servo.h>

Servo myServo;

// Pins
const int servoPin = 9;
const int trigPin = 10;
const int echoPin = 11;

// Adjust these based on your servo direction
const int closePos = 0;   // Lid down
const int openPos = 180;   // Lid open (adjust as needed)

const int openThreshold = 20; // cm

bool isOpen = false;

void setup() {
  myServo.attach(servoPin);
  myServo.write(closePos); // Start closed
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
  Serial.begin(9600);
}

void loop() {
  int distance = measureDistance();

  Serial.print("Distance: ");
  Serial.print(distance);
  Serial.println(" cm");

  if (distance > 0 && distance < openThreshold) {
    if (!isOpen) {
      openTrash();
      isOpen = true;
    }
  } else {
```

```
    if (isOpen) {
      closeTrash();
      isOpen = false;
    }
  }

  delay(200);
}

int measureDistance() {
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);

  long duration = pulseIn(echoPin, HIGH, 30000); // 30ms timeout
  if (duration == 0) return -1;

  int distance = duration * 0.034 / 2;
  return distance;
}

void openTrash() {
  for (int pos = closePos; pos <= openPos; pos += 2) {
    myServo.write(pos);
    delay(10);
  }
}

void closeTrash() {
  for (int pos = openPos; pos >= closePos; pos -= 2) {
    myServo.write(pos);
    delay(10);
  }
}
```

Listing 3: Final Arduino Code for Smart Dustbin

Working Principle

The ultrasonic sensor detects the presence of a hand or object near the bin. When the distance is less than a defined threshold (e.g., 20 cm), the servo motor rotates to lift the lid open. Once the object moves away and the distance exceeds the threshold, the servo returns to its original position, closing the lid.

This process is continuously repeated, enabling real-time interaction without physical contact. The bin acts as a basic real-time responsive system using embedded sensors and actuators.

4 Conclusion

The development of a Smart Garbage Bin Monitoring System presents a meaningful contribution to the growing field of IoT and smart city applications. This project not only enhances public hygiene by eliminating the need for physical contact with waste bins, but also introduces a smarter, automated solution for urban waste management. By leveraging ultrasonic sensing and servo motor actuation, the system enables real-time detection of user presence and efficient control of the bin lid.

Furthermore,This project aligns directly with Saudi Arabia's Vision 2030 by promoting sustainable infrastructure, digital transformation, and improved quality of life in urban environments like Jeddah and Riyadh. It also showcases how embedded systems and IoT can be combined to address local challenges using accessible, low-cost hardware like Arduino and basic sensors.

overall, this system not only offers a practical, real-world solution to improve public cleanliness, but also serves as a foundational example of how simple technologies can drive smarter, more efficient cities. With future upgrades, such as wireless communication or AI-based fill prediction, the smart bin system has the potential to become a key component in the broader smart city ecosystem.

5 References

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